

1. It's generally agreed that when you throw a fair die the probability that you roll a multiple of 3 is $\frac{1}{3}$.

I roll what I know to be a fair die twenty times without getting a multiple of 3.

Is the probability that I will roll a multiple of 3 on my twenty-first roll

- (a) considerably greater than $\frac{1}{3}$,
- (b) approximately equal to $\frac{1}{3}$,
- (c) considerably less than $\frac{1}{3}$?

Explain your answer.

2. I have a list of hard mental arithmetic questions. When I ask my pupils these questions, about a third of the questions are answered correctly.

I have asked Treehorn twenty of these questions. He has got none of them right.

Is the probability that Treehorn gets the next question I ask him correct

- (a) considerably greater than $\frac{1}{3}$,
- (b) approximately equal to $\frac{1}{3}$,
- (c) considerably less than $\frac{1}{3}$?

Explain your answer.

3. William's Latin teacher gives him a test approximately one in every three Latin lessons.

His Latin teacher has not given him a test for twenty lessons in a row.

Is the probability that there will be a test in his next Latin lesson

- (a) considerably greater than $\frac{1}{3}$,
- (b) approximately equal to $\frac{1}{3}$,
- (c) considerably less than $\frac{1}{3}$?

Explain your answer.

4. I overheard the following in Sainsbury's Local. A lady was chatting to the checkout assistance next to the lottery machine:

"Some people say that the more tickets you buy, the more chance you have of getting the winner, but for me it doesn't work like that. You've either got the winner or you haven't. It doesn't matter whether you bought one tickets or hundreds."

The following is in a text about medical diagnosis:

"When a patient consults his physician with an undiagnosed disease, neither he nor the doctor knows whether it is rare until the diagnosis is finally made. Statistical methods can only be applied to a population of thousands. The individual either has a rare disease or doesn't have it; the relative incidence of two diseases is completely irrelevant to the problem of making his diagnosis".

Do you think probability is being discussed correctly in these statements? Do you think either person has any valid points?

5. Linda is 31 years old, single, outspoken and very bright. She has a first-class degree in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.

Rank the following statements in order from most likely, to least likely (in your opinion):

- a. Linda is a teacher in elementary school.
 - b. Linda works in a bookstore and takes Yoga classes.
 - c. Linda is active in the feminist movement.
 - d. Linda is a psychiatric social worker.
 - e. Linda is a member of the League of Women voters.
 - f. Linda is a bank teller.
 - g. Linda is an insurance salesperson.
 - h. Linda is a bank teller and is active in the feminist movement.
6. Imagine that you play a game on your mobile phone which requires considerable investment of time and/or money. After playing it for over a year, with perhaps a few hundred pounds invested, you are contacted by someone who tells you that if you pay them £50 of your own money, then they will load your account with credits which can be used to purchase £1000 worth of items. This is against the terms of service of the app, however, and if you are caught doing this then your account will be banned.

If you were made this offer would you accept it, if you believe the probability of being caught and banned is:

- | | | |
|---------------|---------|--------|
| a. 0.00000001 | b. 0.01 | c. 0.1 |
| d. 0.49 | e. 0.51 | f. 0.8 |
7. A person contacts IT support having a problem with their phone. For some reason their email has stopped syncing to their phone. They have tried re-entering all of their credentials, but it still isn't working. The IT helper believes that there is a 0.95 probability that the mail app will need to have its updates removed and be restored to factory condition, and then the updates will need to be reinstalled before linking the account to the phone. There is also a 0.05 probability that the problem will be fixed by turning the phone off and on again. What should the IT helper suggest?
8. A dentist discovers that patient has a deep cavity, and decides that a normal filling might not be successful, and instead a root canal might be required. The dentist can perform the filling today, but if it is unsuccessful then within a few days the patient will be in a very large amount of pain (and will still require a trip to a dentist for a root canal). On the other hand, the patient has an expensive overseas family holiday booked from tomorrow, which will have to be cancelled if a root canal is performed (because a root canal requires several visits to the dentist to complete).

If you were the patient, what would you decide to do if the dentist tells you that the probability that a filling will be unsuccessful is:

- | | | |
|---------|--------|---------|
| a. 0.05 | b. 0.3 | c. 0.48 |
| d. 0.52 | e. 0.7 | f. 0.95 |